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1. (a) Let $G = \{e, (12)(34), (1234)(56), (13)(24), (1432)(56), (56)(13), (14)(23), (24)(56)\}$

- i. Find the stabilizer and the orbit of 3.

$$\begin{aligned} \text{stab}_G(3) &= \{e, (24)(56)\} \\ \text{orb}_G(3) &= \{3, 4, 1, 2\} \end{aligned}$$

- ii. Find the stabilizer and the orbit of 5.

$$\begin{aligned} \text{stab}_G(5) &= \{e, (12)(34), (13)(24), (14)(23)\} \\ \text{orb}_G(5) &= \{5, 6\} \end{aligned}$$

- (b) Calculate the orders of the following groups.

- i. The group of rotations of a regular octahedron (a solid with eight congruent equilateral triangles as faces).

$$* |S| = 8$$

For some face f , it can be mapped to every side of the octahedron. So,

$$|\text{orb}_G(i)| = 8.$$

Also, since each face is a triangle, 3 rotations will leave f in the same place. That is,

$$|\text{stab}_G(i)| = 3.$$

By the Orbit Stabilizer Theorem, $|G| = |\text{stab}_G(i)| \cdot |\text{orb}_G(i)| = 8 \cdot 3 = \boxed{24}$

- ii. The group of rotations of a regular dodecahedron (a solid with 12 congruent regular pentagons as faces).

$$* |S| = 12$$

Recall from class that a group of rotations on a soccer ball, when looking at the black spots (pentagons), also has a set S of order 12. Since the same is true here, $|G|$ must be the same as the group of rotations on a soccer ball. That is,

$$\begin{aligned} |\text{orb}_G(i)| &= 12 \\ |\text{stab}_G(i)| &= 5. \end{aligned}$$

By the Orbit Stabilizer Theorem,

$$|G| = |\text{orb}_G(i)| \cdot |\text{stab}_G(i)| = 12 \cdot 5 = \boxed{60}$$

2. (a) Let G be a group of permutations of a finite set S . Show that, for any $i \in S$, $\text{stab}_G(i)$ is a subgroup of G .

Proof. Let G be a group of permutations of a finite set S . It must be nonempty since the identity permutation fixes any element i to itself. So, $e \in \text{stab}_G(i)$. Now, let $\alpha, \beta \in \text{stab}_G(i)$. This means α and β keep i fixed. By the Finite Subgroup Test, we need to show that $\alpha\beta \in \text{stab}_G(i) \forall \alpha, \beta \in \text{stab}_G(i)$. Using the definition of composition,

$$\begin{aligned}\alpha\beta(i) &= \alpha(\beta(i)) \\ &= \alpha(i) \\ &= i.\end{aligned}$$

i also stays fixed in $\alpha\beta$, making $\alpha\beta \in \text{stab}_G(i)$. Since the elements of G permute a finite number of elements, G itself must be finite. Thus, this is enough to show that $\text{stab}_G(i) \leq G$ by the Finite Subgroup Test. $\square$

(b) Prove that the external direct product of any finite number of groups is a group.

Proof. To show that $G_1 \oplus G_2 \oplus \dots \oplus G_n$ is a group, it must satisfy the defining properties of groups.

Closure: Let $t_1 = (g_1, g_2, \dots, g_n)$ and $t_2 = (h_1, h_2, \dots, h_n)$ where $g_i, h_i \in G_i$ and $t_1, t_2 \in G_1 \oplus G_2 \oplus \dots \oplus G_n$. We want to show that $t_1 t_2 \in G_1 \oplus G_2 \oplus \dots \oplus G_n$.

$$\begin{aligned}t_1 t_2 &= (g_1, g_2, \dots, g_n)(h_1, h_2, \dots, h_n) \\ &= (g_1 h_1, g_2 h_2, \dots, g_n h_n) \text{ where } g_i h_i \in G_i \text{ (by closure)} \\ &\in (G_1 \oplus G_2 \oplus \dots \oplus G_n).\end{aligned}$$

This shows that componentwise operations are indeed a binary operation on E.D.Ps.

Associative: This is inherent since componentwise operations are associative.

Identity: Let e be the n -tuple of identity elements and $t = (g_1, g_2, \dots, g_n) \in G_1 \oplus G_2 \oplus \dots \oplus G_n$. Then,

$$\begin{aligned}t_1 e &= (g_1, g_2, \dots, g_n)(e_1, e_2, \dots, e_n) \text{ where } e_i \in G_i \\ &= (g_1 e_1, g_2 e_2, \dots, g_n e_n) \\ &= (g_1, g_2, \dots, g_n).\end{aligned}$$

So, $e \in G_1 \oplus G_2 \oplus \dots \oplus G_n$.

Inverse: Let $t_1 = (g_1, g_2, \dots, g_n)$ and $t_1^{-1} = (g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$. Thus,

$$\begin{aligned}t_1 t_1^{-1} &= (g_1, g_2, \dots, g_n)(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1}) \\ &= (g_1 g_1^{-1}, g_2 g_2^{-1}, \dots, g_n g_n^{-1}) \\ &= (e, e, \dots, e) \\ &= e.\end{aligned}$$

So, the external direct product of any finite number of groups is a group. $\square$

3. (a) Show that $G \oplus H$ is Abelian if and only if G and H are Abelian.

Proof. ($\Rightarrow$) Suppose $G \oplus H$ is Abelian. That means

$$\Rightarrow (g_1g_2, h_1h_2) = (g_2g_1, h_2h_1) \text{ where } g_1, g_2 \in G, h_1, h_2 \in H.$$

Since the operation of E.D.Ps are componentwise, this equality implies

$$g_1g_2 = g_2g_1 \text{ and } h_1h_2 = h_2h_1.$$

This is precisely the definition of commutativity between elements of G and H , making the groups Abelian.

($\Leftarrow$) Suppose G and H are Abelian groups. This means

$$\begin{aligned} g_1g_2 &= g_2g_1 \quad \forall g_i, g_j \in G \\ h_1h_2 &= h_2h_1 \quad \forall h_i, h_j \in H \\ \Rightarrow (g_1g_2, h_1h_2) &= (g_2g_1, h_2h_1). \end{aligned}$$

Since both tuples contain elements from G and H , both tuples are elements of $G \oplus H$. So, $G \oplus H$ is Abelian. $\square$

(b) The dihedral group of D_n of order $2n$ ($n \geq 3$) has a subgroup of order n (of all the rotations) and many subgroups of order 2 (e.g $\langle F_1 \rangle$). Explain why D_n cannot be isomorphic to the external direct product of two such groups.

Consider R to be any subgroup of rotations of order n and F to be any subgroup of order 2. By the Cayley Table from HW 3, both of these groups are cyclic, and thus Abelian. By 3(a), that makes $R \oplus F$ Abelian. However, D_n for $n \geq 3$ is not Abelian. So, in general, D_n can **not** be isomorphic to the external direct product of such groups.

4. (a) Prove, by comparing orders of elements, that $\mathbb{Z}_8 \oplus \mathbb{Z}_2$ is not isomorphic to $\mathbb{Z}_4 \oplus \mathbb{Z}_4$.

$$* \mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}, \mathbb{Z}_2 = \{0, 1\}, \mathbb{Z}_4 = \{0, 1, 2, 3\}$$

$$\begin{aligned} \mathbb{Z}_8 \oplus \mathbb{Z}_2 = \{ & (0,0), (0,1), \\ & (1,0), (1,1), \\ & (2,0), (2,1), \\ & (3,0), (3,1), \\ & (4,0), (4,1), \\ & (5,0), (5,1), \\ & (6,0), (6,1), \\ & (7,0), (7,1) \} \end{aligned}$$

$$\begin{aligned} \mathbb{Z}_4 \oplus \mathbb{Z}_4 = \{ & (0,0), (0,1), (0,2), (0,3), \\ & (1,0), (1,1), (1,2), (1,3), \\ & (2,0), (2,1), (2,2), (2,3), \\ & (3,0), (3,1), (3,2), (3,3) \} \end{aligned}$$

By Theorem 8.1, consider the order of $(1,1)$ in $\mathbb{Z}_8 \oplus \mathbb{Z}_2$,

$$\begin{aligned} |(1_{\mathbb{Z}_8}, 1_{\mathbb{Z}_2})| &= \text{lcm}(|1_{\mathbb{Z}_8}|, |1_{\mathbb{Z}_2}|) \\ &= \text{lcm}(8, 2) \\ &= 8. \end{aligned}$$

If $\mathbb{Z}_8 \oplus \mathbb{Z}_2 \cong \mathbb{Z}_4 \oplus \mathbb{Z}_4$, there must exist some element in $\mathbb{Z}_4 \oplus \mathbb{Z}_4$ of order 8. However,

$$\left. \begin{array}{l} |0| = 1 \\ |1| = 4 \\ |2| = 2 \\ |3| = 4 \end{array} \right\} \text{orders in } \mathbb{Z}_4$$

it is impossible to get the form $\text{lcm}(|a|, |b|) = 8$ where $a, b \in \mathbb{Z}_4$ from these orders. So, $\mathbb{Z}_4 \oplus \mathbb{Z}_4$ does **not** have an element of order 8 and hence $\mathbb{Z}_8 \oplus \mathbb{Z}_2 \not\cong \mathbb{Z}_4 \oplus \mathbb{Z}_4$.

(b) Prove that $\mathbb{Z} \oplus \mathbb{Z}$ is not cyclic.

Proof. By contradiction, suppose $\mathbb{Z} \oplus \mathbb{Z}$ is cyclic. That means $\exists$ some tuple $(a, b) \in \mathbb{Z} \oplus \mathbb{Z}$ s.t $\langle (a, b) \rangle = \mathbb{Z} \oplus \mathbb{Z}$. Thus, $\forall (c, d) \in \mathbb{Z} \oplus \mathbb{Z}$, there's some $n \in \mathbb{Z}$ s.t

$$(c, d) = \underbrace{(a, b) + (a, b) + \dots + (a, b)}_{n \text{ times}} = (na, nb).$$

However, consider the elements $(1, 0), (0, 1) \in \mathbb{Z} \oplus \mathbb{Z}$. If $\mathbb{Z} \oplus \mathbb{Z}$ is cyclic, $(1, 0), (0, 1) \in \langle (a, b) \rangle$. Let $x, y \in \mathbb{Z}$ s.t

$$(xa, xb) = (1, 0) \quad \text{and} \quad (ya, yb) = (0, 1).$$

Since the operation of E.D.Ps are componentwise,

$$\Rightarrow xa = 1 \text{ and } xb = 0 \quad yb = 1 \text{ and } ya = 0$$

$$\Rightarrow x = a = \pm 1 \text{ and } \underline{b = 0} \quad y = b = \pm 1 \text{ and } \underline{a = 0}.$$

So, the only way to generate both $(1, 0)$ and $(0, 1)$ is if $(a, b) = (0, 0)$. But, $\langle (0, 0) \rangle = \{(0, 0)\}$. Thus, $\mathbb{Z} \oplus \mathbb{Z}$ is not cyclic. $\square$

(c) Adapt your proof from part (b) to prove that, for any nontrivial group G , $\mathbb{Z} \oplus G$ is not cyclic.

Proof. By contradiction, suppose $\mathbb{Z} \oplus G$ is cyclic. That means $\exists$ some $a \in \mathbb{Z}$ and $g \in G$ s.t $\langle (a, g) \rangle = \mathbb{Z} \oplus G$. $\forall (c, d) \in \mathbb{Z} \oplus G$ where $c \in \mathbb{Z}$ and $h \in G$, there's some $n \in \mathbb{Z}$ such that

$$(c, d) = (a, g)^n = (na, g^n).$$

Similar to (a), consider the elements $(1, e), (0, g') \in \mathbb{Z} \oplus G$ where $g' \neq e$. Let $x, y \in \mathbb{Z}$ s.t

$$(xa, g^x) = (1, e) \quad \text{and} \quad (ya, g^y) = (0, g').$$

Since the operation of E.D.Ps are componentwise,

$$\Rightarrow xa = 1 \text{ and } g^x = e \quad ya = 0 \text{ and } g^y = g'$$

$$\Rightarrow x = a = \pm 1 \text{ and } \underline{g^1 = g^{-1} = e} \quad \underline{y = 0} \text{ and } \underline{g^0 = g' = e}.$$

So, the only way to generate both $(1, e)$ and $(0, g')$ is if $(a, g) = (0, e)$. But, if e is a generator of G , that contradicts G being nontrivial. Thus, $\mathbb{Z} \oplus G$ is not cyclic. $\square$